

1. Is $f(x) = x^3 - x$ invertible or not? Justify your answer.

Solution. Since $f(0) = 0$ and also $f(1) = 0$, f fails the horizontal line test. That is, f is not one-to-one. Since f is not one-to-one, it is not invertible.

2. Let $h(x) = x^7 + 3x^3 + 1$. Compute $(h^{-1})'(5)$.

Solution. It looks like it will be difficult to actually invert h , so instead we use the inverse function theorem: $(h^{-1})'(5) = 1/h'(h^{-1}(5))$. Thus we need to know the derivative of h and the value of h^{-1} at 5. We can do the first task with the power rule:

$$h'(x) = 7x^6 + 9x^2$$

and the second task by guess-and-check. We find that $h(1) = 1^7 + 3 \cdot 1^3 + 1 = 1 + 3 + 1 = 5$, so $h^{-1}(5) = 1$. Thus

$$(h^{-1})'(5) = \frac{1}{h'(1)} = \frac{1}{7 \cdot 1^6 + 9 \cdot 1^2} = \frac{1}{16}.$$

3. Compute $\frac{d}{dx} \cos(\ln|x-3|)$.

Solution. We have

$$\begin{aligned} \frac{d}{dx} \cos(\ln|x-3|) &= -\sin(\ln|x-3|) \cdot \frac{d}{dx} \ln|x-3| \\ &= -\sin(\ln|x-3|) \cdot \frac{1}{x-3} = \frac{-\sin(\ln|x-3|)}{x-3} \end{aligned}$$

applying the chain rule twice.

4. Compute $\frac{d}{dx} x^{(e^x)}$.

Solution. We're trying to differentiate a function that consists of one function raised to the power of another, so we use logarithmic differentiation. Setting $y = x^{e^x}$ and applying the natural log function on both sides gives $\ln(y) = e^x \ln(x)$. Differentiating both sides gives

$$\frac{1}{y} \cdot y' = e^x \ln(x) + e^x \cdot \frac{1}{x},$$

which we can solve for y' to get

$$\frac{d}{dx} x^{e^x} = y' = y \left(e^x \ln(x) + \frac{e^x}{x} \right) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right).$$